

Language understanding involves translating natural language inputs—which can be ambiguous, elliptical, ill-formed, and implicature-laden—into fully specified, ontologically grounded text meaning representations (TMRs) that convey their intended meaning. The global interpretation of an input is built up from the interpretations of progressively larger groups of words, phrases, and constructions, informed by features of the context.

Language generation, for its part, involves converting ontologically grounded generation meaning representations (GMRs), which result from an agent's reasoning about action, into contextually appropriate natural language utterances.

As described earlier (section 2.1.2), TMRs and GMRs are very similar apart from their metadata. Moreover, many of same linguistic and extralinguistic challenges confront both language understanding and generation, albeit in different guises. For example:

- The language understander must disambiguate polysemous words, whereas the language generator must select among available paraphrases.
- The language understander must resolve referring expressions whereas the language generator must choose among options for referring to entities.
- The language understander must reconstruct elided material, whereas the language generator must decide when to produce elliptical utterances.
- The language understander must interpret indirect speech acts, whereas the language generator must decide when to use them.

Language understanding and generation mirror each other on the input and output flanks of the agent architecture, as shown by figure 4.1, which is a language-specific version of the generic architecture in figure 2.1.

The language-oriented path through the architecture differs from the generic architecture as follows:

- The input is speech or text.
- The generic module Perception Recognition is realized as Language Recognition.
- The generic module Perception Interpretation is realized as Language Understanding, which yields a text meaning representation (TMR).
- Action Specification yields a GMR (generation meaning representation), which is specific to language generation.

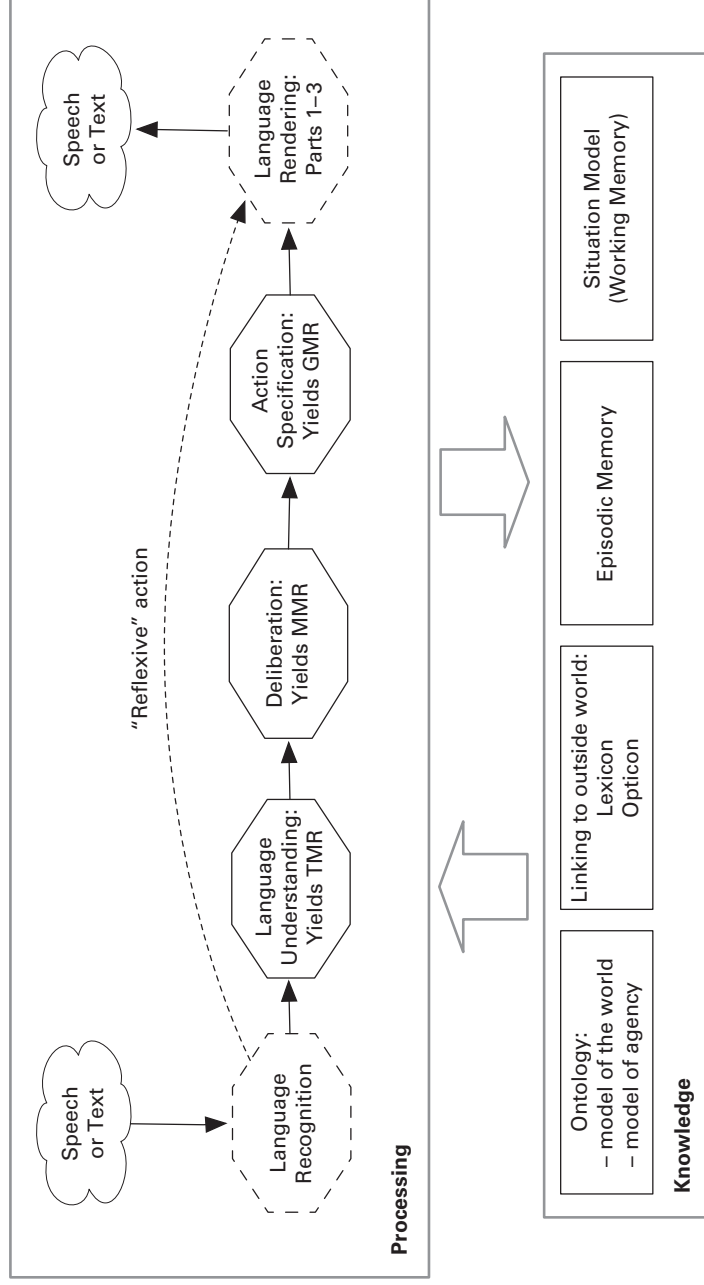


Figure 4.1
Natural language understanding and generation reflect a specialization of the generic architecture shown in figure 2.1.

- The generic module Action Rendering is realized as Language Rendering, which is a three-part process.
- The output is speech or text.

Language understanding and generation are modeled as hybrid processes, meaning that they benefit from the contributions of data-driven tools, indicated by the dotted borders on two of the system modules in figure 4.1. We will discuss language understanding and generation in turn.

4.2 Language Understanding

This section begins with a high-level recap of the approach to natural language understanding reported in *Linguistics for the Age of AI* and then reports recent advances in the microtheory of construction semantics.

4.2.1 Brief Overview of Language Understanding

The input to language understanding is a text, possibly transcribed from speech, which undergoes a maximum of six stages of analysis. This staging reflects both principles of cognitive modeling and the demands of system building. The six stages, which are detailed in chapters 3–7 of *Linguistics for the Age of AI*, are:

Stage 1. Basic Syntax runs external, data-driven tools within a custom wrapper. Collected features are primarily morphological and syntactic.¹ Using data-driven tools for morphological and syntactic analysis is practical because: (1) the theory of Ontological Semantics (Nirenburg & Raskin, 2004) makes no claims about how humans compute morphology and syntax; (2) the mainstream natural language processing community has spent tremendous resources on morphology and syntax over the past half century, and the resulting tools produce useful results; and (3) morphological and syntactic analysis do not need to be perfect since LEIAs interpret the associated features as overridable heuristic evidence.

Stage 2. OntoSyntax uses the features collected in Basic Syntax to begin populating the data structure that will ultimately hold the text meaning representation (TMR). The agent also carries out extensive additional analysis of the input, which involves such things as selecting which lexical senses might be used to analyze the input, treating syntactic transformations, and initiating the learning of new word senses.

Stage 3. **Basic Semantics** focuses on lexical disambiguation and the establishment of the semantic dependency structure for individual sentences. Typically, this stage produces multiple candidate TMRs, which are improved upon and, ideally, pruned down to a single one by the end of stage 6.

Stage 4. **Lexically Triggered Procedural Semantics** runs specialized routines that are called from the meaning-procedures zone of lexical senses. These cover many different kinds of phenomena, from coreference (*she*) to scalar modifiers (*very hot*) to words whose meanings guide the interpretation of a whole utterance (*respectively*).

Stage 5. **Extended Semantics** attempts to resolve residual ambiguities, incongruities, and underspecifications using:

- a. additional knowledge bases, such as ontologically grounded repositories for analyzing nominal compounds and metonymies;
- b. additional algorithms, such as the ones that attempt to specify ungrounded comparisons (*X is better*—than what?) and understand indirect modification (*a fast road* is a road where one *can drive* fast); and
- c. a multi-sentence window of discourse, along with analysis methods that exploit it.

Stage 6. **Situational Reasoning** invokes all of an agent's abilities—such as plan-and-goal-oriented reasoning, image analysis, and mindreading of the interlocutor—in an attempt to arrive at a single, precise, contextually appropriate, high-confidence analysis of a language input.

Each stage of language analysis results in a knowledge structure that the agent can use to reason about *actionability*. It asks itself, *Is the nascent TMR, in its current state, adequate input to reasoning about action?* For example, a LEIA that is tasked with fundamentally analyzing only inputs about a particular topic can, after only one or two stages of analysis, determine whether the input contains words or concepts of interest; if not, its action can be to ignore it. Such skimming is useful for learning by reading and for agents exposed to off-topic chitchat by their human collaborators. Similarly, an agent tasked with learning new words by reading might focus on the results of Basic Semantics, opting to exclude any sentences that require coreference resolution or other advanced reasoning.²

One way to think about language understanding is in terms of knowledge used and knowledge produced. LEIAs use both prerecorded (static) knowledge and knowledge generated during system operation. This reflects the following principle of content-centric cognitive modeling: decision heuristics